

$$2.(2) \quad \lim_{n \rightarrow \infty} f_n(x) = x^2,$$

$$|f_n(x) - f(x)| = \sqrt{x^4 + e^{-n}} - x^2 = \frac{e^{-n}}{\sqrt{x^4 + e^{-n}} + x^2} < e^{-n} \rightarrow 0$$

故一致收敛

$$(5) \quad \lim_{n \rightarrow \infty} f_n(x) = 0$$

$$(1) \quad |f_n(x) - f(x)| = \frac{\sin x^n}{3+x^n} \leq \frac{x^n}{3} \leq \frac{(1-\delta)^n}{3}, \text{ 而 } 1-\delta \in (0,1), \text{ 故一致收敛}$$

$$(2) \quad \text{取 } x_n = \sqrt[n]{\frac{n-1}{n}}$$

$$|f_n(x_n) - f(x_n)| = \frac{\sin(1-\frac{1}{n})}{4-\frac{1}{n}}, \quad \lim_{n \rightarrow \infty} |f_n(x_n) - f(x_n)| = \frac{\sin 1}{4}, \text{ 故不一致收敛}$$

$$3.(2) \quad S_n = x - \frac{x^{n+1}}{n+1}$$

$$\lim_{n \rightarrow \infty} S_n = x, \text{ 则 } S(x) = x$$

$$|S_n(x) - S(x)| = \left| \frac{x^{n+1}}{n+1} \right| \leq \frac{1}{n+1} \rightarrow 0,$$

故部分和序列 $\{S_n\}$ 一致收敛

故级数一致收敛

$$(6) \quad \sin x \cdot \sin nx = -\frac{1}{2} (\cos(n+1)x - \cos(n-1)x)$$

$$\text{故 } \sum_{n=1}^{\infty} \sin x \sin nx = -\frac{1}{2} (\cos 2x - \cos 0 + \cos 2x - \cos 4x + \dots + \cos(n+1)x - \cos(n-1)x)$$

$$= -\frac{1}{2} (\cos(n+1)x + \cos nx - \cos x - 1) \leq 2$$

故 $\sum_{n=1}^{\infty} \sin x \sin nx$ 一致有界

$$\text{令 } B_n = \frac{1}{n^2+x^2}, \quad B_n(x) \text{ 关于 } n \text{ 单调}, \quad \lim_{n \rightarrow \infty} B_n(x) = 0$$

$$|B_n(x) - B(x)| = \frac{1}{n^2+x^2} \leq \frac{1}{n^2} \rightarrow 0$$

故 $\{B_n\}$ 一致收敛于 0

由狄利克雷判别法, 原级数一致收敛

5. 证明: 级数

$$g(x) = \sum_{n=1}^{\infty} 2^n \sin \frac{x}{3^n}$$

在 $(-\infty, +\infty)$ 中不一致收敛, 但在任意闭区间 $[-M, M]$ ($M > 0$) 上一致收敛, 并证明 $g(x)$ 在 $(-\infty, +\infty)$ 中有连续的导函数.

$$u_n = 2^n \cdot \sin \frac{x}{3^n}, \text{ 取 } x_n = 3^n.$$

$$u_n(x_n) = 2^n \cdot \sin 1, \quad \lim_{n \rightarrow \infty} u_n(x_n) \neq 0, \text{ 故不一致收敛}$$

当 $x \in [-M, M]$ 时

$$\left| 2^n \sin \frac{x}{3^n} \right| \leq \left| \left(\frac{2}{3}\right)^n \cdot x \right| \leq |M| \left(\frac{2}{3}\right)^n, \text{ 而 } \sum_{n=1}^{\infty} M \left(\frac{2}{3}\right)^n \text{ 收敛}$$

故 $g(x)$ 在 $[-M, M]$ 上一致收敛

$$g'(x) = \left(\frac{2}{3}\right)^n \cdot \cos \frac{x}{3^n}, \text{ 在 } (-\infty, +\infty) \text{ 上连续}$$

$$\text{而 } \left| \left(\frac{2}{3}\right)^n \cos \left(\frac{x}{3^n}\right) \right| \leq \left(\frac{2}{3}\right)^n,$$

故 $g'(x)$ 在 $(-\infty, +\infty)$ 上一致收敛

故 $g(x)$ 在 $(-\infty, +\infty)$ 中有连续导函数

8. 设 $\sum_{n=1}^{\infty} a_n$ 收敛. 证明级数 $\sum_{n=1}^{\infty} \frac{a_n}{n^x}$ 在 $[0, +\infty)$ 中一致收敛, 并有

$$\lim_{x \rightarrow 0+0} \sum_{n=1}^{\infty} \frac{a_n}{n^x} = \sum_{n=1}^{\infty} a_n.$$

(提示: 注意 $\sum_{n=1}^{\infty} \frac{a_n}{n^x}$ 在 $[0, +\infty)$ 中的连续性.)

$\sum_{n=1}^{\infty} a_n$ 对任意 x 一致收敛

而序列 $\left\{ \frac{1}{n^x} \right\}$ 对 $x \in [0, +\infty)$ 单增, 且 $\frac{1}{n^x} \leq 1$, 一致有界

由阿贝尔判别法, $\sum_{n=1}^{\infty} \frac{a_n}{n^x}$ 在 $[0, +\infty)$ 上一致收敛.

而任取 n , $u_n(x) = \frac{a_n}{n^x}$ 连续

$$\text{故 } \lim_{x \rightarrow 0+0} \sum_{n=1}^{\infty} \frac{a_n}{n^x} = \sum_{n=1}^{\infty} \lim_{x \rightarrow 0+0} \frac{a_n}{n^x} = \sum_{n=1}^{\infty} a_n$$

$$10, 5 \quad 2, (3) \quad \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{(2n+1) \cdot (2n+1)!}{(2n+3)(2n+3)!} x^2 = \lim_{n \rightarrow \infty} \frac{2n+1}{(2n+2)(2n+3)^2} x^2 = 0$$

$\therefore R = +\infty$, 收敛区间与域均为 $\mathbb{R}$

$$(6) a_n = 3^{-n} + 5^{-n} = 5^{-n} \cdot \left(\left(\frac{3}{5}\right)^n + 1\right) = \frac{1}{5^n} \left[\left(\frac{3}{5}\right)^n + 1\right] = \frac{1}{3^n} \left[\left(\frac{3}{5}\right)^n + 1\right]$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \frac{1}{3} \quad \therefore R = 3$$

$x=3$ 时, 易知发散; $x=-3$ 时, $u_n = (-1)^n + (-1)^n \cdot \left(\frac{3}{5}\right)^n$, $\lim_{n \rightarrow \infty} |u_n| = 1$, 发散

$\therefore$ 收敛区间均为 $(-3, 3)$

$$(9) \quad \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^{n+1} \frac{1}{k}}{\sum_{k=1}^n \frac{1}{k}} = 1, \quad x = \pm 1 \text{ 时均发散} \quad \text{故收敛区间/域为 } (-1, 1)$$

$$(10) \quad \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(2n+2)!!}{(2n+3)!!} \cdot \frac{(2n+1)!!}{(2n)!!} = \lim_{n \rightarrow \infty} \frac{2n+2}{2n+3} = 1$$

$$\textcircled{1} x=1 \text{ 时}, \quad \therefore \frac{n-1}{n} < \frac{n}{n+1} \quad \frac{n}{n-1} > \frac{n+1}{n}$$

$$\therefore \frac{2 \times 4 \times \cdots \times 2n}{1 \times 3 \times 5 \times \cdots \times (2n+1)} > \frac{1}{2n+1} \cdot \frac{3 \times 5 \times \cdots \times (2n+1)}{2 \times 4 \times \cdots \times 2n} = \frac{1}{2n+1} \cdot \frac{(2n+1)!!}{(2n)!!}$$

$$\therefore u_n^2 > \frac{1}{2n+1}, \quad u_n > \frac{1}{\sqrt{2n+1}}, \quad \therefore \text{发散.}$$

$$\textcircled{2} x=-1 \text{ 时}, \quad \therefore \frac{(2n)!!}{(2n+1)!!} < \frac{(2n+1)!!}{(2n)!!} \cdot \frac{1}{2n}, \quad \therefore |u_n| < \frac{1}{\sqrt{2n}}, \quad \therefore u_n \rightarrow 0.$$

又有 $|u_n| \downarrow$

故收敛

$\therefore$ 收敛域为 $[-1, 1)$, 区间为 $(-1, 1)$

$$3. (4) \quad S(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n(2n-1)} x^{2n}, \quad R=1, \quad -1 < x < 1$$

$$S'(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} \cdot 2x^{2n-1}, \quad S''(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \cdot 2 \cdot x^{2n-2} = 2 \sum_{n=1}^{\infty} (-1)^{n-1} x^{2n-2} = \frac{2}{1+x^2}$$

$$S'(0) = 0, \quad S'(x) = 2 \arctan x$$

$$S(0) = 0, \quad S(x) = 2 \int_0^x \arctan t \, dt = 2 \cdot \left(x \arctan x - \int_0^x \frac{x}{1+x^2} dx \right) = 2x \arctan x - \ln(x^2+1)$$

$$(6) \quad R = \sqrt{2}, \quad x \in (-\sqrt{2}, \sqrt{2}) \quad \left(\sum_{n=1}^{\infty} \frac{x^{2n-1}}{2^n} \right)', \quad \sum_{n=1}^{\infty} \frac{x^{2n-1}}{2^n} = \lim_{n \rightarrow \infty} \frac{\frac{x}{2} (1 - (\frac{x^2}{2})^n)}{1 - \frac{x^2}{2}} = \frac{x}{2-x^2}$$

$$S(0) = 0, \quad \therefore S(x) = \int_0^x \frac{t}{2-t^2} dt = -\frac{1}{2} \ln |2-x^2|$$